Bayesian Optimization of Multi-Material 3D-Printed Tensegrity Structures for Energy Absorption

BYU Ira A. Fulton College of Engineering
Mentored Research Grant Proposal

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Co-PI: Sterling G. Baird, Department of Mechanical Engineering

Duration: 2 Years — **Requested:** $25,000

Abstract

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize multi-material 3D-printed tensegrity structures for energy absorption. Tensegrity structures—assemblies of rigid compression members (PLA) and flexible tension members (TPU)—offer tunable mechanical responses and high strength-to-weight ratios. Using a multi-fidelity Bayesian optimization framework, students will combine finite element simulations (low-fidelity) with physical experiments (high-fidelity) to efficiently navigate the high-dimensional design space. Students will gain hands-on experience in CAD design, multi-material 3D printing, impact testing, computational modeling, and data-driven optimization. Expected outcomes include undergraduate conference presentations (UCUR, ASME IDETC), a peer-reviewed journal submission, and development of marketable technical skills. This project provides a rich mentored learning environment at the intersection of structural mechanics, advanced manufacturing, and machine learning.

Number of undergraduates: 2–3
Number of graduate students: 1 (co-mentor)

Budget Summary

Category	**Year 1**	**Year 2**	**Total**
Undergraduate wages	$7,500	$7,500	$15,000
Graduate student wages	$2,000	$2,000	$4,000
Supplies	$2,750	$750	$3,500
Travel	$0	$2,000	$2,000
Other	$250	$250	$500
Total	**$12,500**	**$12,500**	**$25,000**

Relationship to External Funding

The PIs do not currently hold external funding for tensegrity or architected materials research. This MRG provides the primary support for establishing a mentored undergraduate research program in this area. If the program proves successful, the preliminary data and mentoring framework developed here may inform a future NSF proposal; however, the MRG is self-contained and its outcomes—trained student researchers, conference presentations, and a journal submission—do not depend on external funding.

1 Research Motivation and Overview

We propose a two-year mentored research program in which 2–3 undergraduate students design and optimize **multi-material 3D-printed tensegrity structures** for energy absorption. Tensegrity (tensional integrity) structures—assemblies of rigid compression members suspended within a continuous network of tension members—exhibit remarkable strength-to-weight ratios and tunable mechanical responses, making them attractive for protective equipment, packaging, and aerospace applications [Skelton and de Oliveira, 2009, Fraternali et al., 2015].

Recent advances in multi-material 3D printing enable the fabrication of tensegrity-like structures on a single print platform, using PLA for rigid struts and TPU for flexible tension elements. This eliminates tedious manual assembly and opens a rich, high-dimensional design space. Navigating this space efficiently requires intelligent experimental design; **multifidelity Bayesian optimization** (BO) provides exactly this capability by combining inexpensive simulations (low-fidelity) with physical experiments (high-fidelity) to guide the search toward optimal configurations [Mo et al., 2023, Perdikaris et al., 2017].

The primary objective of this project is to provide a rich, interdisciplinary mentored research experience for undergraduates. Students will gain hands-on experience in CAD design, multi-material 3D printing, impact testing, computational modeling, and data-driven optimization—skills directly transferable to graduate school and engineering careers. Each student will lead a clearly scoped research project, present at conferences, and contribute to a peer-reviewed publication.

Figure placeholder: Overview of project approach—tensegrity unit cell design (left), multi-material 3D printing of PLA struts and TPU tension elements (center), and multifidelity Bayesian optimization loop connecting simulations to experiments (right).

Figure 1: Conceptual overview of the proposed research and mentoring framework.

2 Background

Tensegrity structures consist of isolated compression members (bars or struts) held in stable equilibrium by a continuous network of tension members (cables or strings) [Skelton and de Oliveira, 2009, Sultan, 2009]. Their defining characteristic—that no two rigid bars touch—yields lightweight assemblies with surprising load-bearing capacity, resilience, and tunable nonlinear responses [Amendola et al., 2014, Zhang et al., 2015]. Prior work has demonstrated their potential for impact protection [Fraternali et al., 2015], but design optimization remains largely unexplored due to the high-dimensional parameter space (bar/cable lengths, thicknesses, stiffnesses, topology, and geometry).

Bayesian optimization is a sequential, model-based strategy for optimizing expensive-to-evaluate black-box functions [Shahriari et al., 2016, Frazier, 2018]. A Gaussian process surrogate model is fitted to observed data, and an acquisition function balances exploration with exploitation. BO has been applied to materials discovery [Lookman et al., 2019] and metamaterial design [Wang et al., 2022]. Critically, Mo et al. [Mo et al., 2023] recently demonstrated **multifidelity BO** for architected materials, using inexpensive simulations as a low-fidelity source and physical experiments as high-fidelity data. We adopt this framework directly: finite element simulations serve as the low-fidelity source, and 3D-printed tensegrity impact tests provide high-fidelity validation.

3 Student Research Project 1: Simulation and Bayesian Optimization

Student profile: One junior or senior undergraduate with coursework in mechanics of materials or computational methods.

Scope and Deliverables

This student will develop and run the computational pipeline:

1. **Parameterize a tensegrity unit cell** suitable for simulation and multi-material 3D printing, including strut geometry (PLA), tension element geometry (TPU), topology, and boundary conditions.
2. **Build finite element models** using ABAQUS or open-source FEA tools (e.g., FEniCS) to predict force–displacement response and energy absorption under compaction and drop-test loading. These simulations serve as the *low-fidelity* source in the multifidelity framework.
3. **Implement a multifidelity Bayesian optimization loop** (following Mo et al. [Mo et al., 2023]) using Python libraries (BoTorch/Ax) to efficiently explore the design space, fusing simulation predictions with experimental results from Student 2.
4. **Identify optimal designs** and generate predictions for experimental validation.

Mentoring outcomes: The student will develop skills in finite element analysis, scientific computing (Python), and data-driven optimization. They will present results at **ASME IDETC** or **UCUR** and contribute as co-author on a journal manuscript.

4 Student Research Project 2: Fabrication, CAD, and Experimental Testing

Student profile: One junior or senior undergraduate with coursework in machine design, CAD, or compliant mechanisms.

Scope and Deliverables

This student will lead fabrication and physical testing:

1. **Design tensegrity structures in CAD** (SolidWorks or Fusion 360) for multi-material 3D printing, with PLA struts and TPU tension elements printed on a single platform.
2. **Fabricate structures** using the department's multi-material 3D printer, iterating on print parameters to achieve reliable tensegrity geometries.

3. **Conduct experimental testing:** compaction tests (quasi-static compression), drop tests (dynamic impact), and wave propagation measurements (hitting on one end and measuring time response on the other). These experiments serve as the *high-fidelity* source.
4. **Compare experimental results** against simulation predictions from Student 1, quantifying model accuracy and informing surrogate updates.

Mentoring outcomes: The student will develop skills in CAD, additive manufacturing, test design, and data analysis. They will present at **UCUR** or **ASME IDETC** and contribute as co-author on a journal manuscript.

5 Mentoring Environment

The mentoring of undergraduate researchers is the central objective of this proposal. We are committed to providing a structured, supportive environment that develops students into independent, capable researchers.

Recruitment

Students will be recruited from courses in mechanics of materials, machine design, kinematics, and compliant mechanisms through brief in-class presentations. We target junior or senior undergraduates. We will actively seek to recruit students from underrepresented groups in engineering.

Mentoring Structure

- **Weekly individual meetings** (30 min) with faculty PI (Prof. Hill) and Co-PI (Prof. Baird) for technical guidance, troubleshooting, and professional development discussions.
- **Weekly lab group meetings** (1 hour) including: progress updates, student presentations on recent literature, career discussions, and skill-building workshops (e.g., scientific writing, conference presentation skills, research ethics).
- **Graduate student co-mentor:** Prof. Baird's master's student will serve as day-to-day co-mentor, providing hands-on guidance in simulation, coding, fabrication, and experimental technique. This layered mentoring model ensures students receive timely support while the graduate student develops their own mentoring skills.
- **Progressive responsibility:** Students begin with structured tasks (running simulations with provided scripts, assembling structures from detailed CAD plans) and progress to independent design decisions, culminating in ownership of their research project.
- **Peer mentoring:** In Year 2, experienced students will mentor newly recruited freshmen or sophomores, reinforcing their own learning while expanding the research group and creating a sustainable pipeline of trained researchers.

Student Development Outcomes

Students will develop a broad, transferable skill set spanning:

- **Technical:** experimental mechanics (test design, data acquisition), computation (FEA modeling, Python, Bayesian optimization), and design (CAD, multi-material 3D printing).

- **Communication:** lab notebooks, group presentations, conference talks, and co-authoring a peer-reviewed manuscript.
- **Professional:** project management, collaboration with interdisciplinary teams, and preparation for graduate school or industry careers.

6 Expected Research Outcomes

1. **Trained undergraduate researchers:** 2–3 students with hands-on experience in advanced manufacturing, computational modeling, and data-driven optimization, prepared for graduate study or engineering careers.
2. **Conference presentations:** Student-led presentations at **UCUR** (Utah Conference on Undergraduate Research) and **ASME IDETC** (International Design Engineering Technical Conferences).
3. **Journal submission:** One peer-reviewed manuscript submitted to the *ASME Journal of Mechanical Design* or *Smart Materials and Structures*, with undergraduate students as co-authors.
4. **Open-source tools:** Simulation code, optimization scripts, and experimental data published on GitHub for reproducibility.

7 Potential Impact of Work

This project advances the design of architected materials for energy absorption with direct applications in **protective equipment** (helmets, body armor), **packaging** (shock-absorbing inserts), and **aerospace** (lightweight impact-resistant structures). By demonstrating that multifidelity Bayesian optimization can efficiently navigate the tensegrity design space, we establish a generalizable framework applicable to other classes of architected materials and metamaterials.

The interdisciplinary training—spanning structural mechanics, advanced manufacturing, and machine learning—prepares students for graduate study or careers in industries increasingly reliant on data-driven design. The open-source dissemination of tools and data extends impact beyond BYU and enables other research groups to build on our work.

8 Timeline

9 Budget

Total requested: **$25,000** over 2 years. Per grant guidelines, no funds are allocated to equipment, tuition, or faculty wages.

Budget Justification

Undergraduate wages ($15,000): The largest budget item supports 2–3 undergraduate research assistants across the two-year project. Wages provide meaningful compensation for the

Task	Y1 Fall	Y1 Winter	Y2 Fall	Y2 Winter
Student recruitment & onboarding	•			
Literature review	•	•		
Tensegrity parameterization & CAD	•	•		
FEA simulation development		•	•	
Multifidelity BO implementation		•	•	
Multi-material 3D printing & fabrication		•	•	•
Impact testing & data collection			•	•
Simulation–experiment comparison				•
Conference presentations (UCUR/IDETC)			•	•
Journal manuscript preparation				•
Peer mentoring of new students			•	•

Table 1: Project timeline across four semesters.

Category	Description	Year 1	Year 2
Undergraduate wages	2–3 students, hourly research assistantship	$7,500	$7,500
Graduate student wages	Partial RA for graduate co-mentor	$2,000	$2,000
Supplies	Filament, load cells, DAQ, fasteners, compute	$2,750	$750
Travel	Student conference registration & travel	$0	$2,000
Other	Poster printing, lab consumables, contingency	$250	$250
Annual Total		**$12,500**	**$12,500**

Table 2: Proposed budget summary ($25,000 total over 2 years).

significant time commitment required for simulation development, multi-material 3D printing, and experimental testing campaigns.

Graduate student wages ($4,000): A small allocation supports Prof. Baird's master's student who serves as a day-to-day co-mentor, providing hands-on guidance in simulation, coding, fabrication, and experimental technique, amplifying the mentoring capacity of the faculty PIs.

Supplies ($3,500): Includes 3D printing filament (PLA for struts, TPU for tension elements), assembly hardware (fasteners, jigs, adhesives), miniature load cells, a USB-based data acquisition module, and cloud computing credits for simulation campaigns. Heavier procurement occurs in Year 1 during initial setup.

Travel ($2,000): Supports undergraduate student presentations at ASME IDETC or UCUR in Year 2, covering registration, transportation, and lodging.

Other ($500): Covers poster printing, lab consumables, and unforeseen expenses.

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Biographical Sketches

Jeffrey R. Hill — Principal Investigator

Professional Preparation

Ph.D.	Mechanical Engineering	University of Utah
M.S.	Mechanical Engineering	Brigham Young University
B.S.	Mechanical Engineering	Brigham Young University

Appointments

- Associate Professor, Department of Mechanical Engineering, Brigham Young University (current)
- Research Engineer, Sandia National Laboratories (prior)

Products Most Closely Related to This Proposal

Publications and products related to smart materials, shape memory alloys, compliant mechanisms, and structural mechanics.

Synergistic Activities

- Mentoring of undergraduate and graduate students in experimental mechanics and smart materials research at BYU.
- Teaching courses in mechanics of materials, machine design, and kinematics in the Department of Mechanical Engineering.
- Collaboration with national laboratories on advanced materials characterization and structural design.

Sterling G. Baird — Co-Principal Investigator

Professional Preparation

Ph.D.	Materials Science and Engineering	University of Utah
B.S.	Materials Science and Engineering	Brigham Young University

Appointments

- Faculty, Department of Mechanical Engineering, Brigham Young University (current)

Products Most Closely Related to This Proposal

Publications and products related to materials informatics, Bayesian optimization, data-driven materials discovery, and machine learning for materials science.

Synergistic Activities

- Development of open-source tools for Bayesian optimization and materials informatics.
- Mentoring of graduate and undergraduate students in data-driven methods for materials science and engineering.
- Active contributions to the materials informatics community through publications, software, and educational resources.